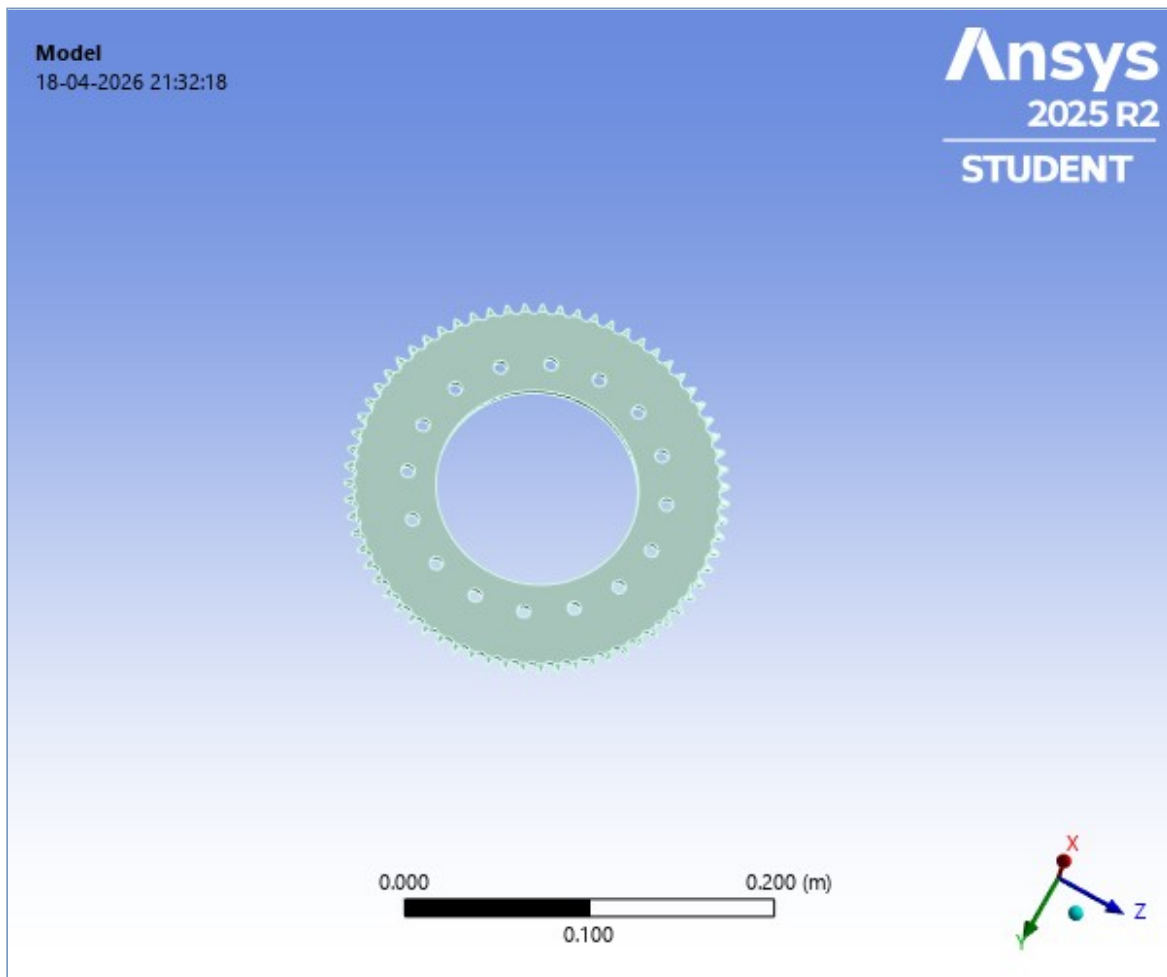




Project

First Saved	Saturday, October 18, 2025
Last Saved	Friday, November 28, 2025
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4)

TABLE 2

Model (A4, B4) > Geometry Imports	
Object Name	Geometry Imports
State	Solved

TABLE 3

Model (A4, B4) > Geometry Imports > Geometry Import (A3, B3)

--	--

Object Name	<i>Geometry Import (A3, B3)</i>
State	Solved
Definition	
Source	C:\Users\samar\weightReduction_files\dp0\SYS\DM\SYS.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4, B4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\samar\weightReduction_files\dp0\SYS\DM\SYS.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	5.3196e-003 m
Length Y	0.20867 m
Length Z	0.20872 m
Properties	
Volume	1.1381e-004 m ³
Mass	0.89339 kg
Scale Factor Value	1.
Statistics	
Bodies	1

Active Bodies	1
Nodes	55086
Elements	29110
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4) > Geometry > Parts

Object Name	SYS\ DE 35
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	5.3196e-003 m
Length Y	0.20867 m

Length Z	0.20872 m
Properties	
Volume	1.1381e-004 m ³
Mass	0.89339 kg
Centroid X	2.6598e-003 m
Centroid Y	-2.9694e-007 m
Centroid Z	5.2634e-007 m
Moment of Inertia Ip1	5.9026e-003 kg·m ²
Moment of Inertia Ip2	2.9534e-003 kg·m ²
Moment of Inertia Ip3	2.9534e-003 kg·m ²
Statistics	
Nodes	55086
Elements	29110
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:175.159.143	

TABLE 6
Model (A4, B4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (A4, B4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined

Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (A4, B4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	7.3797e-004 m
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	0
Active Connections	0

Mesh

TABLE 10
Model (A4, B4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	

Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	2.e-003 m
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	0.29519 m
Average Surface Area	1.1179e-004 m ²
Minimum Edge Length	3.8399e-006 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	2
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	55086
Elements	29110
Show Detailed Statistics	No

Named Selections

TABLE 11
Model (A4, B4) > Named Selections > Named Selections

Model (A4, B4) - Named Selections - Named Selections		
Object Name	<i>Outer</i>	<i>Inner</i>
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	

Geometry	55 Faces	16 Faces
Definition		
Send to Solver	Yes	
Protected	Program Controlled	
Visible	Yes	
Program Controlled Inflation	Exclude	
Statistics		
Type	Manual	
Total Selection	55 Faces	16 Faces
Surface Area	1.3731e-003 m²	1.0016e-003 m²
Suppressed	0	
Used by Mesh Worksheet	No	

Static Structural (A5)

TABLE 12
Model (A4, B4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 13
Model (A4, B4) > Static Structural (A5) > Analysis Settings

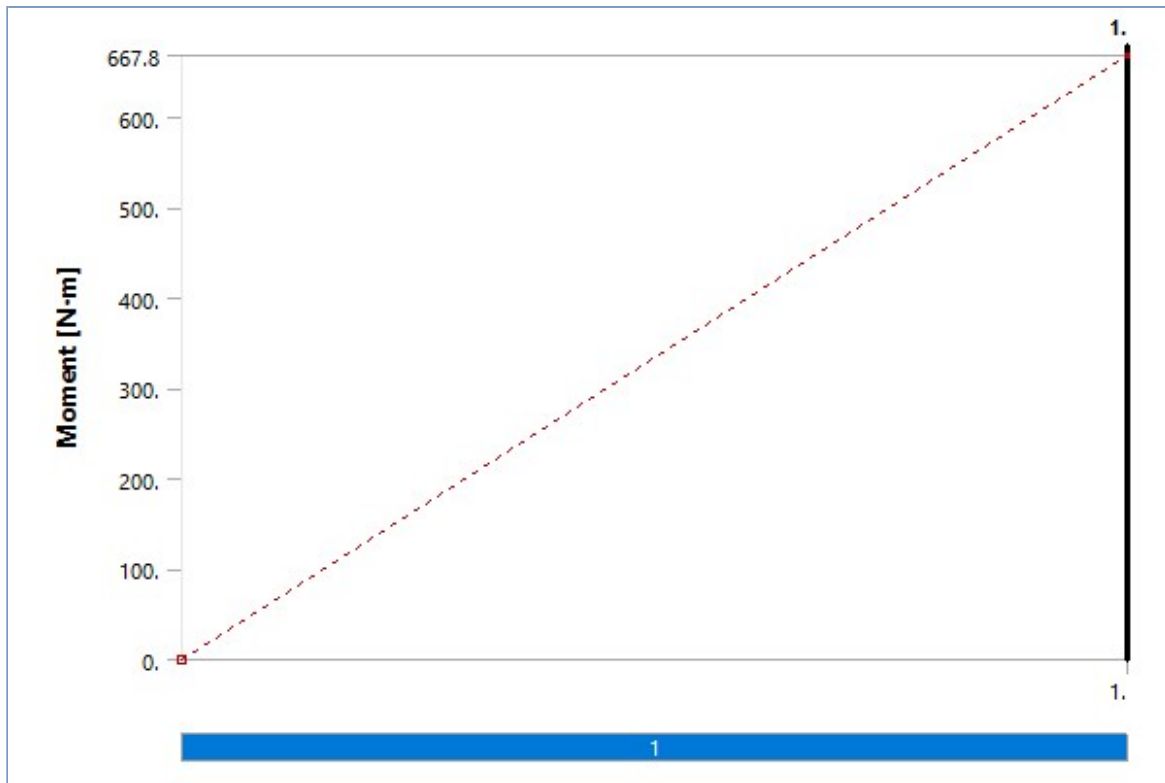
Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled

Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	Yes
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\samar\weightReduction_files\dp0\SYS\MECH\
Future Analysis	Structural Optimization
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 14
Model (A4, B4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Moment
State	Fully Defined	
Scope		
Scoping Method	Named Selection	
Named Selection	Outer	Inner
Definition		
Type	Fixed Support	Moment
Suppressed	No	
Define By		Vector
Magnitude		667.8 N·m (ramped)
Direction		Defined
Behavior		Deformable
Advanced		
Pinball Region		All

FIGURE 1
Model (A4, B4) > Static Structural (A5) > Moment



Solution (A6)

TABLE 15
Model (A4, B4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	18. s
MAPDL Memory Used	900. MB
MAPDL Result File Size	33.5 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 16
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Solution Information

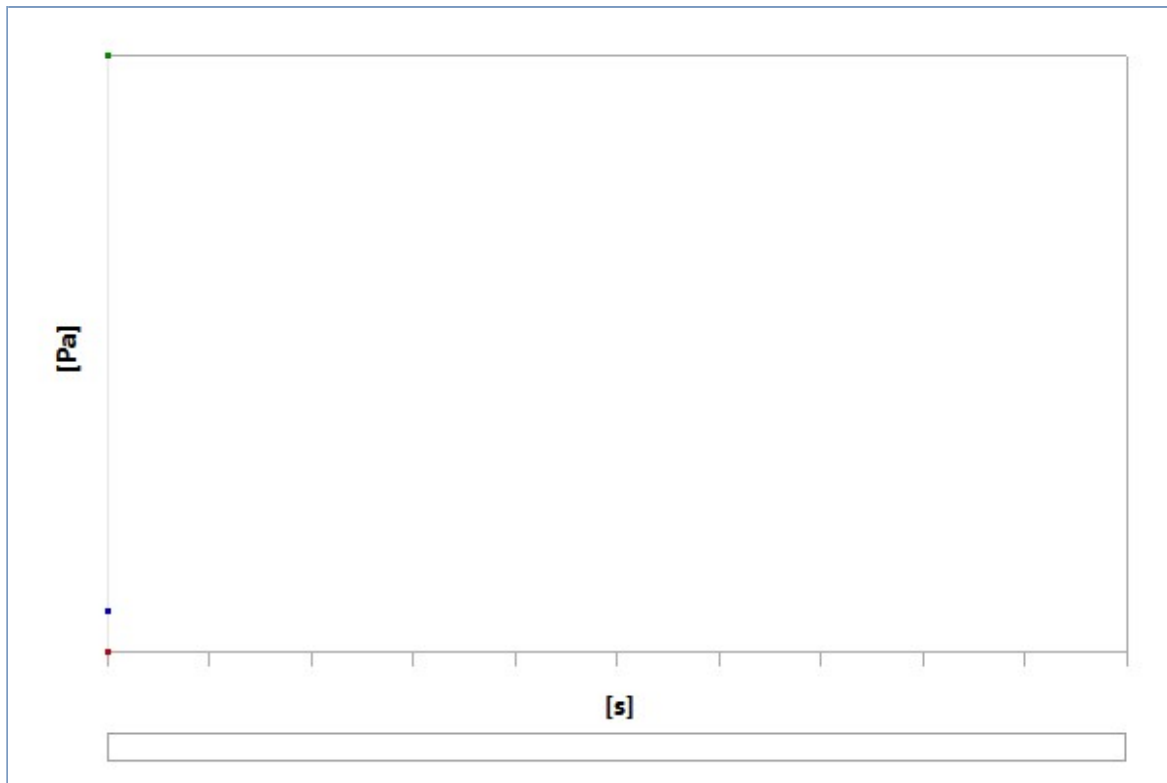
Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes

Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Equivalent Stress	Total Deformation
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Total Deformation
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	720.97 Pa	0. m
Maximum	7.916e+007 Pa	1.0913e-005 m
Average	5.5073e+006 Pa	4.0763e-006 m
Minimum Occurs On	SYS\ DE 35	
Maximum Occurs On	SYS\ DE 35	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	1	

FIGURE 2
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

**TABLE 18****Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress**

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	720.97	7.916e+007	5.5073e+006

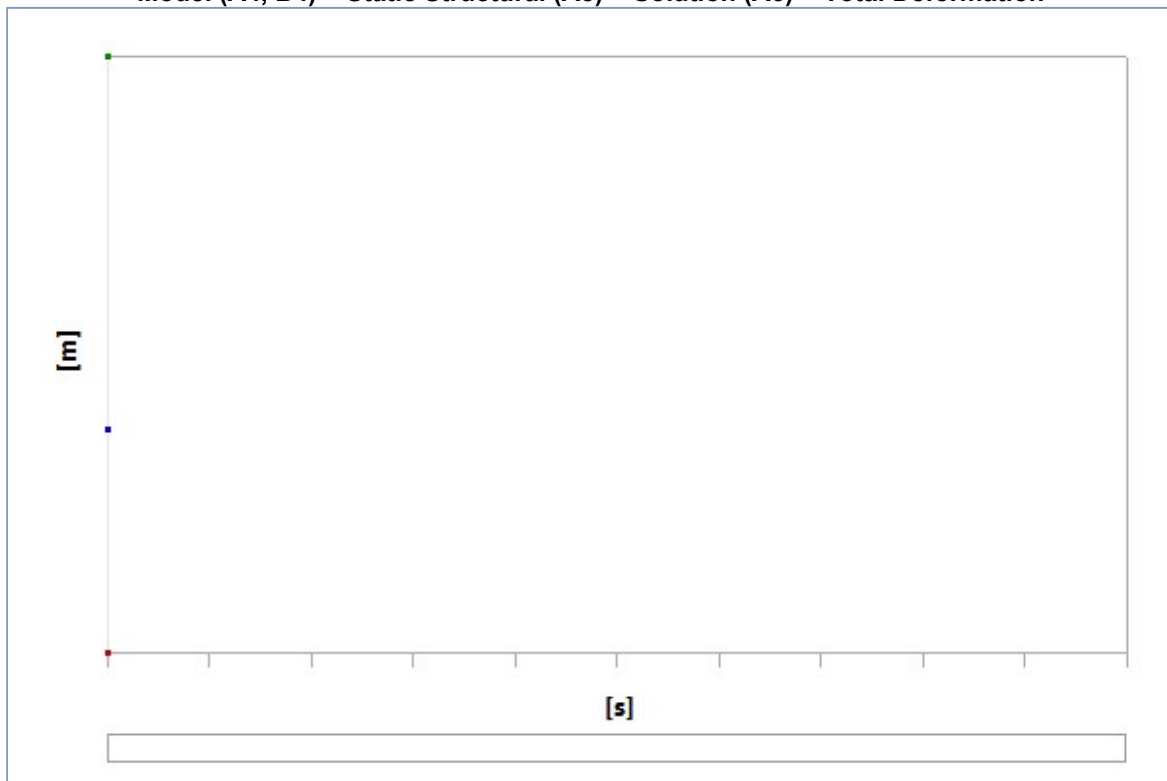
FIGURE 3**Model (A4, B4) > Static Structural (A5) > Solution (A6) > Total Deformation**

TABLE 19
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	1.0913e-005	4.0763e-006

Structural Optimization (B5)

TABLE 20
Model (A4, B4) > Analysis

Object Name	<i>Structural Optimization (B5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Structural Optimization
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 21
Model (A4, B4) > Structural Optimization (B5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Definition	
Maximum Number Of Iterations	500.
Convergence Accuracy	0.1 %
Output Controls	
Export Design Properties	No
Store Results At	All Iterations
Solver Controls	
Solver Type	Program Controlled
Algorithm	Program Controlled
Active Set Strategy	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\samar\weightReduction_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	mks
Max Num Of Intermediate Files	All Iterations

TABLE 22
Model (A4, B4) > Structural Optimization (B5) > Optimization Region

Object Name	<i>Optimization Region</i>
State	Fully Defined
Design Region	
Scoping Method	Geometry Selection
Geometry	All Bodies
Exclusion Region	
Define By	Geometry Selection
Geometry	443 Faces
Exclusion Thickness	Program Controlled
Exclusion Extension	Isotropic

Definition	
Suppressed	No
Optimization Option	
Optimization Type	Topology Optimization - Mixable Density
Initial Volume Fraction	Program Controlled
Penalty Factor (Stiffness)	Program Controlled
Hyperbolic Projection	Program Controlled

TABLE 23
Model (A4, B4) > Structural Optimization (B5) > Objective

Object Name	<i>Objective</i>
State	Fully Defined
Definition	
Suppressed	No
Normalized Sum	No

Model (A4, B4) > Structural Optimization (B5) > Objective

Response Type	Goal	Criterion	Formulation	Environment Name	Weight	Multiple Sets	Start Step	End Step	Step	Start Mode	End Mode	Mode
Compliance	Minimize	N/A	Program Controlled	Static Structural	N/A	Enabled	1	1	1	N/A	N/A	N/A

TABLE 24
Model (A4, B4) > Structural Optimization (B5) > Response Constraint

Model (74, 24) Structural Optimization (25) Response Constraint		
Object Name	Response Constraint	Design Constraint
State	Fully Defined	
Scope		
Scoping Method	Optimization Region	
Optimization Region Selection	Optimization Region	
Definition		
Type	Response Constraint	Design Constraint
Response	Mass	
Define By	Constant	
Percent to Retain	60 %	
Suppressed	No	
Subtype		Cyclic Repetition
Location and Orientation		
Number of Sectors		8
Coordinate System		Global Coordinate System
Axis		X Axis

Solution (B6)

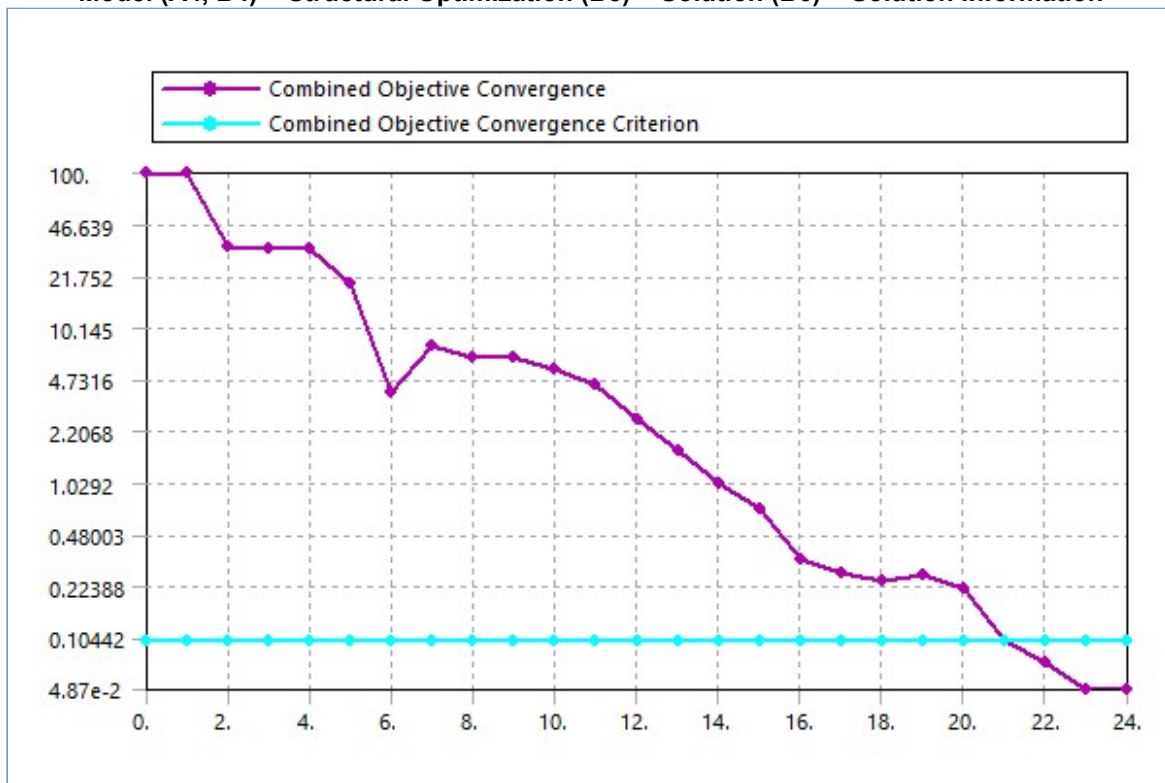
TABLE 25
Model (A4, B4) > Structural Optimization (B5) > Solution

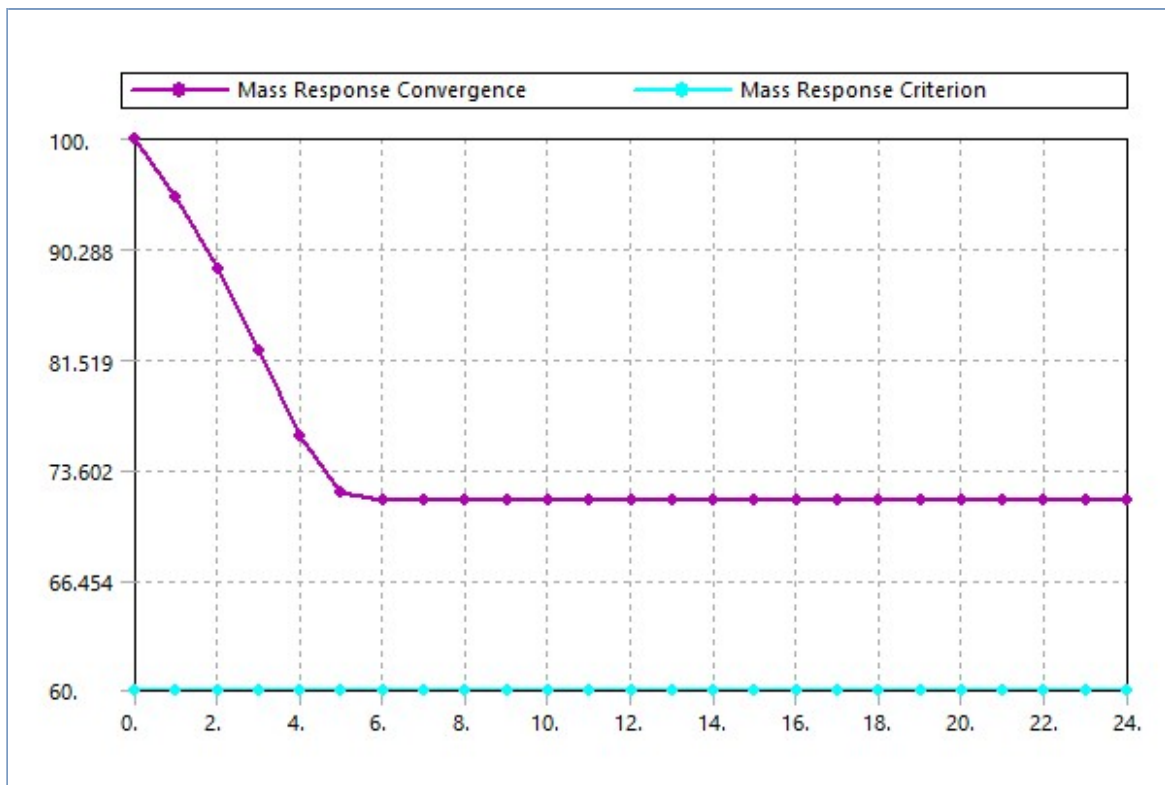
Object Name	<i>Solution (B6)</i>
State	Solved
Information	
Status	Done
MAPDL Elapsed Time	1 m 15 s
MAPDL Memory Used	903. MB
MAPDL Result File Size	1.2115 MB
Post Processing	
Export Optimal Shape	Only Geometry

-- Topology Result	Topology Density
Definition	
Environment Selection List	A5

TABLE 26**Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information**

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Objective & Mass Response Convergence
Response Constraint	Response Constraint
Update Interval	2.5 s
Display Points	All

FIGURE 4**Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information****FIGURE 5****Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information**

**TABLE 27****Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information > Results**

Object Name	<i>Topology Density Tracker</i>
State	Solved
Scope	
Scoping Method	Optimization Region
Optimization Region	Optimization Region
Definition	
Type	Topology Density Tracker
By	Iteration
Iteration	Last
Retained Threshold	0.5
Separate Data by Entity	No
Suppressed	No
Results	
Minimum	0.
Maximum	1.
Average	0.6396
Visibility	
Show Optimized Region	Retained Region
Information	
Iteration Number	23

TABLE 28**Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Results**

Object Name	<i>Topology Density</i>
State	Solved
Scope	
Scoping Method	Optimization Region
Optimization Region	Optimization Region

Definition	
Type	Topology Density
By	Iteration
Iteration	Last
Retained Threshold	0.5
Exclusions Participation	Yes
Separate Data by Entity	No
Calculate Time History	Yes
Suppressed	No
Results	
Minimum	0.
Maximum	1.
Average	0.6396
Original Volume	1.138e-004 m ³
Final Volume	6.7161e-005 m ³
Percent Volume of Original	59.014
Original Mass	0.89336 kg
Final Mass	0.52721 kg
Percent Mass of Original	59.014
Visibility	
Show Optimized Region	Retained Region
Information	
Iteration Number	23

FIGURE 6
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Topology Density

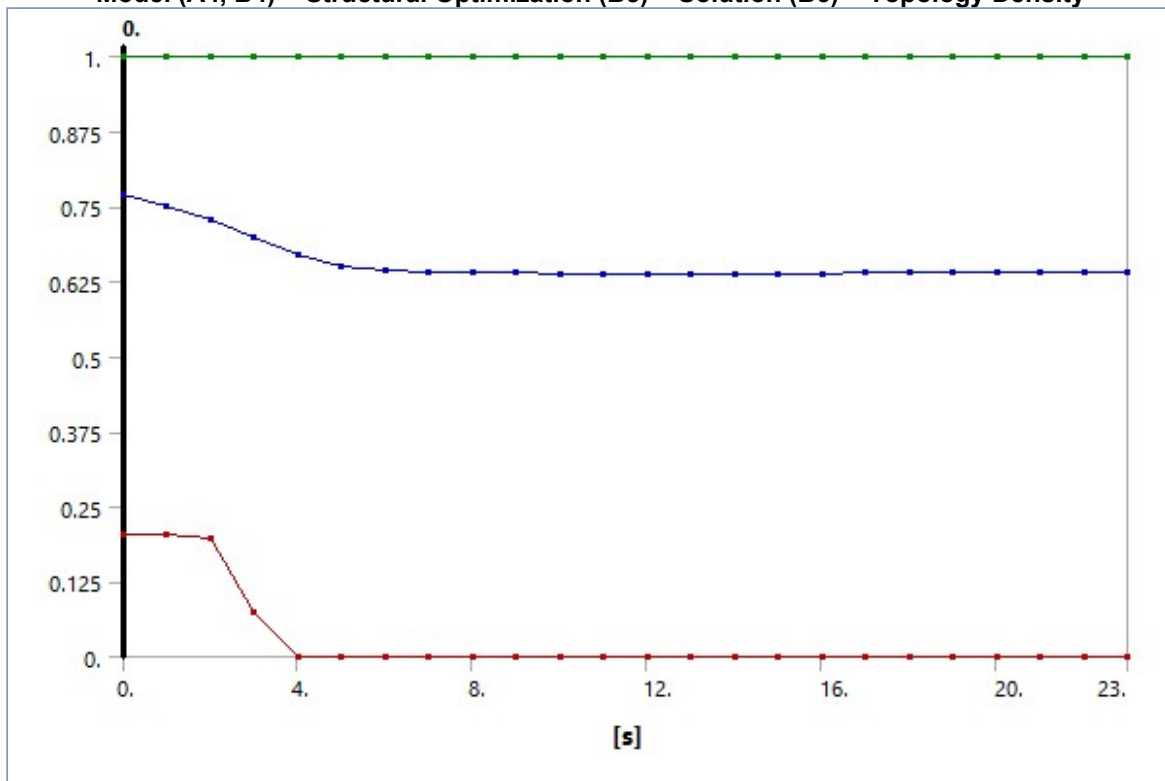


TABLE 29
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Topology Density

Iteration [s]	Minimum	Maximum	Average
0.	0.20272		0.76911

1.	0.20245		0.75091
2.	0.19597		0.72969
3.	7.5991e-002		0.70032
4.	6.2072e-004		0.67007
5.	0.	1.	0.65128
6.			0.64509
7.			0.64233
8.			0.64081
9.			0.63971
10.			0.63891
11.			0.63858
12.			0.6385
13.			0.63865
14.			0.63886
15.			0.63904
16.			0.63914
17.			0.63922
18.			0.63931
19.			0.63943
20.			0.63951
21.			0.63954
22.			0.63958
23.			0.6396

Material Data

Structural Steel

TABLE 30
Structural Steel > Constants

Density	7850 kg m ⁻³
Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat	434 J kg ⁻¹ C ⁻¹
Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Resistivity	1.7e-007 kg m ³ A ⁻² s ⁻³

TABLE 31
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 32
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 33
Structural Steel > Compressive Yield Strength

Compressive Yield Strength Pa
2.5e+008

TABLE 34
Structural Steel > Tensile Yield Strength

Tensile Yield Strength Pa

2.5e+008

TABLE 35
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.6e+008

TABLE 36
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 37
Structural Steel > S-N Curve

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 38
Structural Steel > Strain-Life Parameters

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 39
Structural Steel > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.e+011	0.3	1.6667e+011	7.6923e+010	

TABLE 40
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000